

Vivek Kumar

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EDUCATION

University of Minnesota Twin Cities

Master of Science in Robotics

Minneapolis, MN

Sep 2024 – Present

Relevant Coursework: Machine learning, Computer Vision, Deep Learning, Intelligent Robotic Systems

National Institute of Technology Rourkela

Bachelor of Technology in Mechanical Engineering

Odisha, India

2019 – 2023

SKILLS

Programming & Frameworks: Python, C++, ROS 2, PyTorch, OpenCV, Hugging Face

AI: Vision-Language Models (VLMs), Vision-Language-Action (VLAs), Diffusion Models, Imitation Learning, RL

Robotics & Hardware: SLAM, Path planning, Sensor Fusion, Camera, Lidar, AgileX Rover, AgileX piper arm

Simulation & Tools: Isaac Sim, Gazebo, Rviz, MoveIt, Docker, SolidWorks, Linux

EXPERIENCE

Networking, Mobile, and AI Research Lab (UMN)

Minneapolis, MN

Graduate Research Assistant (*Perception for Autonomous Vehicles*) / Advisor: Prof. Zhi-Li Zhang

Jun 2025 – Present

- Working on implementing and fine-tuning **VLAs** for **end-to-end** autonomous driving, targeting indoor navigation on the AgileX Hunter 2.0 and on-road performance for the Chrysler Pacifica.
- Built a **VLM**-based reasoning framework leveraging Qwen-VL models and NVIDIA Cosmos, fine-tuned on **nuScenes** via **LoRA** and **DPO** alignment to enable proactive scene escalation for complex and long-tail scenarios.
- Developed an **ADAS** suite (FCWS, LDWS, LKAS) using YOLOv8m for **detection**, ByteTrack for **trajectory tracking**, and UFLD-v2 (ResNet-18) for **lane detection**, achieving real-time inference via TensorRT/ONNX acceleration.
- Developed a real time **video prediction** framework by training an encoder-decoder architecture on **BDD100K** dataset to compensate for **100ms+** network latency in systems requiring near-zero latency.
- Implemented a real-time **Bird's-Eye-View (BEV)** pipeline in **ROS 2**, utilizing Inverse Perspective Mapping (IPM), significantly reducing perspective distortion.

GLOVEX Lab (Group of Learning, Optimization, Vision and X)

Minneapolis, MN

Graduate Researcher / Advisor: Prof. Ju Sun

April 2025 – Present

- Researching blind inpainting using unsupervised and **zero-shot** reconstruction methods with pre-trained generative priors based on **diffusion** and **flow-matching** models for image restoration from incomplete observations.
- Designed sparsity-driven **unsupervised** reconstruction frameworks using SCAD loss based optimization and trained domain-specific generative models for robust image restoration and anomaly detection.

PUBLICATIONS

- Vivek Kumar**, et al. *VLM-Driven Proactive Escalation for Tele-operations in Autonomous Vehicles*. **CVPR 2026 Workshop**
- Vivek Kumar**, et al. *Robots That Know When to Ask: VLM-Driven Reasoning for Human-on-Demand Multi-Robot Teleoperation*. **IROS 2026** (Under Review).
- Wenjie Zhang, **Vivek Kumar**, et al. *Zero-Shot Blind Inpainting with Pretrained Diffusion Priors*. **NeurIPS 2026** (Under Review).

PROJECTS

Full-Stack Autonomous Navigation System for AgileX Hunter 2.0

ROS 2, C++, Python

- Developed a production ready modular ROS 2 autonomy stack integrating NDT-OMP LiDAR **localization**, D*+ **path planning**, and collision avoidance for autonomous indoor navigation using RobotSense LiDAR in pre-mapped environments.
- Built the **collision avoidance** pipeline using OBB-voxel intersection tests against **octomap** with automatic replanning capabilities and YAML-based configuration for rapid deployment across different environments.

Multi-Modal Diffusion based Driving Policy

Python, PyTorch

- Developed a lightweight Vision-Language-Action (**VLA**) **diffusion policy** utilizing a custom **CNN-Transformer** architecture to predict autonomous trajectories from natural language instructions.
- Implemented 5-step **Action Chunking** to improve multimodal trajectory prediction and mitigate mode collapse.

Robot Manipulation via Action Chunking Transformers in ManiSkill

Python, PyTorch, ManiSkill

- Built an end-to-end **imitation learning** pipeline using Action Chunking Transformer (ACT) architecture to train robot arm manipulation policy on RGBD observations, achieving **95%+** success rate on PickCube task with 30K training iterations.
- Trained on **HuggingFace** PickCube-v1 dataset, implementing trajectory replay and video-based evaluation pipeline.

On-Demand Mobile EV Charging Robot (SICK LiDAR Competition)

ROS 2, LiDAR, Agile-X

- Worked on an autonomous EV charging-as-a-service platform to mitigate fixed infrastructure barriers and validated it through real-world parking lot testing using a battery-integrated AgileX Hunter platform.